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Mehmet Ali Erdoğan

M.Sc. Student in Computer Engineering

Portfolio:
github.com/MaliErdgn
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M.Sc. student in Computer Engineering specializing in Bioinformatics. Focused on utilizing experience in *System design/development* and *database management* to create innovative solutions in metabolomic data analysis and computational biology.

EDUCATION

M.Sc. in Computer Engineering , Istanbul Technical University	2025 - Present
B.Sc. in Computer Engineering , Namik Kemal University - 3.26	2021 - 2025
B.Sc. in Management Information Systems , Anadolu University (Distance Education) - 3.25	2021 - 2024

PUBLICATIONS

Refereed Journal Publications

- Erdoğan, M. A.**, and Çakmak, A. A High-Dimensional Benchmark of Objective Functions and Biological Resolutions for Personalized Metabolic Phenotyping. *Briefings in Bioinformatics*, 2026 (in press). DOI: [10.1093/bib/bbag528](https://doi.org/10.1093/bib/bbag528).
- Çelik, S., Can, B., **Erdoğan, M. A.**, and Çakmak, A. A Deep Learning Architecture for Combining and Imputing Heterogeneous Metabolomics Datasets. *BMC Bioinformatics*, 2026. DOI: [10.1186/s12859-026-06560-7](https://doi.org/10.1186/s12859-026-06560-7).

Refereed Conference/Workshop Publications

- Şahin, A., **Erdoğan, M. A.**, Kaya, U. S., and Çakmak, A. MetabOmics: Metabolism-Oriented Omics Data Integration. *Proceedings of the ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, June 30 - July 3, 2026, Rende, Italy. DOI: [10.1145/3807503.3819462](https://doi.org/10.1145/3807503.3819462).

Refereed Abstracts and Oral Presentations

- Yiğit, E., Çakmak, A., and **Erdoğan, M. A.** Benchmarking Metabolic Network-based Biomarker Discovery Methods. *Acıbadem Üniversitesi Sağlık Bilimleri Dergisi*, Volume: 17 (Supplement 1), Oral Presentations (to appear).

TECHNICAL EXPERIENCE

Researcher and Teaching Assistant (Turkcell Scholar) / ITU

February 2026 - Present

- Conducting academic research on bioinformatics and metabolic network analysis under the supervision of Assoc. Prof. Ali Çakmak.
- Fulfilling teaching assistant duties for undergraduate Computer Engineering courses and laboratory sessions, and supporting assignment and project evaluation processes.

M.Sc. Project Researcher / TUBITAK

September 2025 - February 2026

- Actively participated in the project titled "(Micro)RNA-Based and Computational Approaches in the Diagnosis, Prognosis, and Treatment of Alzheimer's and Dementia."
- Conducted data analysis and modeling studies using bioinformatics algorithms within the scope of the project.

Intern / COMSIS Bilgisayar Ltd. Sti.

July 2025 - August 2025

- Supported the maintenance and development processes of e-commerce platforms developed using PHP.
- Made user interface (UI) improvements in accordance with web design principles.

Intern / Haroon Technology

March 2025 - May 2025

- Contributed to the development of a PHP-based Customer Relationship Management (CRM) system.
- Wrote database queries using SQL and took part in database management processes.

HONOURS

- Scored 91.54 out of 100 at ALES in 2021.
- Scored 91.25 out of 100 at YÖKDİL at 2023.

SKILLS

Computational Sciences & Data Analysis Bioinformatics & Computational Biology, Machine/Deep Learning (PyTorch, Scikit-learn), Data Analysis (Pandas, NumPy, COBRApy), High-Performance Computing (SLURM)

Programming Languages **Advanced:** Python, JavaScript/TypeScript, SQL, Dart
Intermediate: C, PHP, C#

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Technologies & Frameworks **Web & Backend:** React.js, Next.js, FastAPI, Node.js, Laravel
Mobile: Flutter, React Native

Databases & DevOps **Databases:** PostgreSQL, MongoDB, MySQL, Firebase
DevOps & Infrastructure: Docker (Compose), Nginx, Cloudflare Tunnels, Tailscale, AWS

Tools & Operating Systems Git, Linux (Ubuntu), Windows, LaTeX

PROJECTS

Research & High-Performance Computing (HPC)

- **Metabolomics Data Imputation & ML Pipeline:** Developed deep learning models (including VAEs) for heterogeneous metabolomics datasets. Managed end-to-end experiment pipelines, status tracking, and automated notifications.
- **HPC Cluster Management:** Actively utilized TRUBA and ITU UHEM clusters. Proficient in writing SLURM job scripts, managing PyTorch/CUDA version compatibilities, and executing large-scale training jobs on A100/V100 GPUs.

Cloud Infrastructure & Self-Hosted Deployments

- **Production Server Architecture (*malierdogan.com*):** Migrated and actively maintain all personal and utility web applications on a self-hosted local server infrastructure.
- **DevOps & Networking:** Architected a multi-service containerized environment using Docker Compose. Managed network routing and security using Nginx reverse proxy, automated SSL (Certbot), Cloudflare Tunnels, and Tailscale.
- **Database & Monitoring:** Deployed and maintained remote PostgreSQL and MongoDB instances. Implemented real-time system resource monitoring via Glances.

Full-Stack Web Development

- **Customized AI Chatbot (UN.GPT):** (React, Next.js, AWS, MySQL) Designed and developed an end-to-end AI chatbot interface utilizing the OpenAI API, operating within a restricted knowledge base.
- **Self-Hosted Utility Applications:** Developed and deployed various web utilities to the personal server to demonstrate diverse tech stacks. Includes an API-driven data synchronization platform (**React, FastAPI, PostgreSQL**) and an interactive matchmaking engine with JWT authentication (**Next.js, MongoDB**).

Mobile Application Development

- **Terat - Match Organization Platform:** (Flutter, Dart, Firebase) Developed a comprehensive mobile application utilizing a real-time database that allows users to form teams and organize matches.

ACTIVITIES

Workshops, Seminars, and Conferences

- ACM BCB - ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics - 2026
- TUSEB - Dokuz Eylul University Izmir Genome Workshop - 2025
- HIBIT / RSG - 18th International Symposium on Health Informatics and Bioinformatics - Best Presentation Award - 2025